ASSIGNMENT

What are the differences between **Native**, **Hybrid**, and **Cross-platform** mobile application development, and when is each approach commonly used?

- Native focuses on building an application specifically for one platform, such as iOS or Android, using languages like Swift or Kotlin. Hybrid focuses on building applications using web technologies like HTML, CSS, and JavaScript. Cross-platform, on the other hand, allows developers to build applications that can run on multiple platforms using one codebase, with Flutter as an example.

What is the Flutter Ecosystem, and how does it help developers create modern mobile application?

- It is a collection of tools, libraries, and features that work together with Flutter to help developers build modern mobile applications. It makes development easier by allowing developers to create apps for different platforms using one codebase.

What is the Software Development Life Cycle (SLDC), and why is it important in building successful application?

- It is a step-by-step process used to plan, design, develop, test, and maintain an application. It is important because it helps developers stay organized, reduce errors, and make sure the application meets the user's needs and works properly.

How do flutter development practices and the SLDC work together in creating real-world mobile applications?

- They work together by using Flutter to build the application while following the SDLC steps to guide the development process. Developers plan the app, design its features, build it using Flutter, test it, and then maintain and improve it. This helps create a mobile application that works properly and meets the user's needs.

Why should developers understand these concepts before starting an application project?

- Because it helps them know what tools and processes to use. Understanding Flutter and the SDLC helps developers plan their project properly, avoid mistakes, and create an application that works well and meets the user's needs.